S.Y. B. Tech. Academic Year 2026_27 Semester: III

Data Structures –I Laboratory

LABORATORY WRITE UP

Experiment Number: 04

TITLE: Singly Linked List Operations

PROBLEM STATEMENT:

4. The Department of Computer Engineering is organizing a technical **workshop** for **which** students can **register in** advance. **The registration data of** students **is maintained** using a singly **linked** list. **Each** registered student **is identified by their Registration** Number **(RegNo)** and **Name**. **Write** a C program to **manage the workshop** enrollment system **using** a **singly linked** list. **Implement the** following operations: **a) Add** a **Student** b) Remove a Student c) Display **the** List **d) Count** Total Participants **e) Sort** Lists f**) Reverse the** List

OBJECTIVE:

1. To understand the concept of a singly linked list and dynamic memory allocation using pointers in C.
2. To perform insertion , deletion, display, count nodes, sort, reverse operations on a singly linked list.
3. To develop a menu-driven application for workshop enrollment management.

THEORY

A linked list is a linear data structure in which elements (called nodes) are connected using pointers (or references). Unlike an array, the elements are not stored in contiguous memory**.**

Structure of a Node

Each node contains:

1. Data – the value stored in the node.
2. Pointer (Next) – the address of the next node in the list.

Types of Linked Lists

1. Singly Linked List
    - Each node points to the next node only.
    - Traversal is possible in one direction.
2. Doubly Linked List
    - Each node has two pointers:
        - Previous (prev)
        - Next (next)
    - Traversal is possible in both directions.
3. Circular Linked List
    - The last node points back to the first node.
    - There is no NULL pointer at the end.

Advantages

- Dynamic size (can grow or shrink easily).
- Efficient insertion and deletion (especially at the beginning or in the middle).
- Does not require contiguous memory allocation.

Disadvantages

- Extra memory is needed for pointers.
- Sequential access only

IMPLEMENTATION:

- **PLATFORM:**
    - 64-bit Open source Linux or its derivatives.
    - Open Source C Programming tool like gcc/Eclipse Editor.

- **TEST CONDITIONS:-**

Test Case	Input	Expected Output
TC1	Add one student	Student is added successfully.
TC2	Add multiple students	All students are stored in the linked list.
TC3	Display empty list	Message: "No students registered."
TC4	Display after adding students	All student records are displayed correctly.
TC5	Remove existing student	Student record is deleted successfully.
TC6	Remove non-existing student	Message: "Student not found."
TC7	Count participants	Displays the correct total number of registered students.
TC8	Sort the list	Students are displayed in ascending order of Registration Number.
TC9	Reverse the list	Student records appear in reverse order.
TC10	Delete all students and display	List becomes empty and appropriate message is displayed.

- **PSEUDO CODE:** ***//To be Written by Students***

  Write pseudo code for insertion, deletion, display, count nodes, sort, reverse using Singly Linked List.

- **TIME COMPLEXITY:** ***//To be Written by Students***

  Write time complexity of above operations

- **CONCLUSION:**

  In this practical, a workshop enrollment management system was developed using a singly linked list. The program demonstrates dynamic memory allocation and efficient manipulation of linked data structures. Operations such as insertion, deletion, traversal, counting, sorting, and reversing were successfully implemented. This practical enhances the understanding of linked list operations and their applications in real-world record management systems.

- **FAQs** ***//To be Written by Students***

1. Why is a linked list preferred over an array in this application? What information is stored in each node of the Circular Linked List?
2. What is a singly linked list?
3. What information is stored in each node?
5. What is dynamic memory allocation?
6. Which function is used to release allocated memory?
7. What is the time complexity of inserting a node at the beginning?
8. What is the time complexity of searching for a student?
9. How is the linked list reversed?
10. Which sorting technique can be used in a linked list?

- **PRACTICE ASSIGNMENTS**

1. Develop a C program to manage a workshop enrollment system using a singly linked list. Implement operations to insert student records at the beginning, middle, and end of the list.

2. Modify the workshop enrollment system to search for a student using their Registration Number and display the student's details if found.

3. Develop a program to merge the registration lists of two different workshops into a single linked list.

4. Develop a program to remove duplicate student registrations based on Registration Number.

5. Extend the application by storing additional details such as Department, Year, Email ID, and Mobile Number, and implement all linked list operations on the enhanced student record.